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Software Engineering (60 Minutes)

1. Which is the first phase of the Waterfall software process model?
 1. design
 2. prototype
 3. testing
 4. requirement
2. What is the purpose of use cases in UML?
 1. Requirements capture
 2. Define how the software system will be used
 3. Describe what the user expects to do with the system
 4. Make clear what the stakeholders needs are
3. Please match the Spiral model sectors:

X1: Objective setting
 X2: Risk assessment and reduction
 X3: Development and validation
 X4: Planning

with their correct characteristics:

Y1: Risks are assessed and activities put in place to reduce the key risks
 Y2: Specific objectives for the phase are identified
 Y3: The project is reviewed and the next phase of the spiral is planned
 Y4: A development model for the system is chosen which can be any of the generic models

 1. X1-Y3, X2-Y1, X3-Y2 X4-Y4
 2. X1-Y2, X2-Y3, X3-Y4 X4-Y1
 3. X1-Y2, X2-Y1, X3-Y4 X4-Y3
 4. X1-Y3, X2-Y2, X3-Y1 X4-Y4
4. Indicate what information is provided by Functional requirements?

X1: The constraints on the services or functions offered by the system such as timing constraints.
 X2: How the system should behave in particular situations.
 X3: The constraints on the development process, standards.
 X4: How the system should react to particular inputs.

 1. X2, X4
 2. X1, X2, X4
 3. X1, X3
 4. X2, X3, X4
5. What are the components of a thin client model in Client/Server architecture?
 1. Client (Presentation)-Server (Data Management, Application Processing)
 2. Client (Application Processing)-Server (Data Management)
 3. Client (Data Management)- Server (Application Processing)
 4. Client (Application Processing)- Server-Client (Data Management)
6. Function point is
 1. a pointer to a function
 2. a point where the function is written in a code
 3. a method of estimating the amount of functionality required for a program
 4. a function named "point"
7. A system version
 1. is an instance of system deployed at the client site
 2. is an instance of a system that differs in some way from other instances
 3. should either include new functionalities or should be intended for a different hardware platform
 4. is created to fix reported faults as part of development process
8. What is synchronization control in configuration management?
 1. it governs which software engineer have the authority to access & modify a particular configuration object
 2. it helps to ensure that parallel changes performed by two different people don't overwrite one another
 3. it synchronizes two different system versions to form a single versions
 4. it helps to synchronize the source code files to form deployable version
9. The currently known containment effectiveness of faults introduced during each constructive phase of software development for a particular software product is ratio of
 1. (Actual project duration) to (estimated project duration)
 2. (number of pre release Defects) to (number of pre release Defects + number of post release Defects)
 3. (number of phase i errors) to (number of phase i errors + number of phase i defects)
 4. (number of failure) to (Execution time)
10. SRS is maintained in configuration environment as
 1. Software design baseline
 2. Software development baseline
 3. Software artifacts
 4. Software product baseline
11. Following is the SCM audit tool
 1. Requirement metrics
 2. PERT charts
 3. Source code
 4. Design document
12. Delphi method of cost estimation uses
 1. Functional point analysis
 2. SLOC expressed in KDSI
 3. PERT model using effort calculations
 4. Decomposition method of cost estimation
13. What is 'cleanroom'?
 1. Cleanroom is an area in the software source code, where there are no bugs
 2. Cleanroom is the room requirement for server storage and backup facility, a place where

- temperature is maintained and cleaning is ensured
3. Cleanroom is a software process based on mathematical verification of components and statistical system-level testing
 4. None of the above
14. What is CSCI (recognized through baseline)?
1. Commission for Social Care Inspection
 2. Computer Software Configuration Item
 3. Common Sense Climate Index
 4. Commercial software cost index
15. _____ is a failure if a program doesn't work correctly.
1. Programming
 2. Testing
 3. Both 1 and 2
 4. None of the above
16. Cyclomatic Complexity methods comes under _____ testing method.
1. White box
 2. Black box
 3. Green box
 4. Yellow box
17. A _____ is a sequence of statements from one place in the program to another.
1. Route
 2. Path
 3. Sub path
 4. Gateway
18. Loop Testing comes under which testing method?
1. White Box
 2. Black Box
 3. Green Box
 4. Yellow Box
19. Which of these can be successfully tested using Loop Testing methodology?
1. Simple Loops
 2. Nested Loops
 3. Concatenated Loops
 4. All of the above
20. Graph based testing comes under which testing methods?
1. White Box
 2. Black Box
 3. Green Box
 4. Yellow Box
21. Which testing methods are used by end-users who actually test software before they use it?
1. Alpha and Beta Testing
 2. White Box Testing
 3. Black Box Testing
 4. Trial and Error Testing
22. To test a function, the programmer has to write a _____, which calls the function and passes it test data.
1. Stub
 2. Driver
 3. Proxy
 4. None of the above
23. The configuration verification and audit process:
1. Ensures product quality and that a state of the art design has been achieved
 2. Establishes that the performance and functional requirements defined in configuration documentation have been met
3. Validates the processes used to provide adequate control and visibility
 4. Both 2 and 3
24. What action should management take when a line chart shows a steady variance of project performance from the plan?
1. Adjust the plan accordingly
 2. Do nothing
 3. Accept the deviation as allowable within the scope of the plan
 4. Take immediate action to correct
25. When setting project performance thresholds, overtime is considered. What is the type of threshold?
1. Arbitrary
 2. Management reserve
 3. Target
 4. Statistical
26. The following list of data field items is an example of what type of data integrity problem?
- Developer
Computer Programmer
Programmer
Software Engineer
1. Standardization
 2. Quality
 3. Duplication
 4. Fuzzy logic
27. Using the data from the table below, what is the phase containment of the coding phase?
- | Phase | Number of Defects Introduced | Number of Defects Found & Removed |
|--------------|------------------------------|-----------------------------------|
| Requirements | 12 | 9 |
| Design | 25 | 16 |
| Code | 47 | 42 |
1. 89.4%
 2. 86.4%
 3. 75.0%
 4. 71.2%
28. Software reliability is normally defined in terms of
1. The probability of failure of free operations
 2. The defect density of the software product
 3. The operational profile of the system
 4. Mean time to repair the defect
29. Software defect discovery and removal patterns often follow a profile described by the Rayleigh curve. Which of the following is the most accurate description of this predictive reliability model?
1. More defects discovered and removed early indicates fewer remaining in later phases
 2. Few defects discovered and removed early indicates fewer remaining in later phases
 3. The number of defects discovered and removed early has no effect on reliability
 4. A large number of defects found in later phases indicates errors were injected in later phases
30. Software measurement should be used for all of the following EXCEPT?
1. Predicting reliability of the software in the field
 2. Generating performance reviews of the programmers
 3. Defining the number of software engineers needed

4. Assessing the quality level of the software product
31. The sequential steps for establishing a software quality metrics methodology stated by the IEEE are:
 - I. Identify software quality metrics
 - II. Implement the software quality metrics
 - III. Validate the software quality metrics
 - IV. Establish software quality requirements
 - V. Analyze the software quality metrics
 1. IV, I, II, V, III
 2. I, II, III, IV, V
 3. IV, III, II, I, V
 4. IV, I, II, III, V
32. Software Testing is part of
 1. Quality Control
 2. Quality Assurance
 3. Debugging
 4. None of the above
33. "Degree to which a set of inherent characteristics fulfills requirements." is a definition of
 1. Testing
 2. Quality
 3. Both of the above
 4. None of the above
34. Full form of ISO is
 1. International Organisation for Standardisation
 2. International Standardisation Organisation
 3. International Standards Organisation
 4. None of the above
35. "Management System to direct and control an organization with regard to Quality" is a definition of
 1. Quality
 2. Quality Management System
 3. Corrective Action
 4. Preventive Action
36. "Information and its supporting medium" is a definition of
 1. Record
 2. Document
 3. Evidence
 4. None of the above
37. "Document stating result achieved or providing evidence of activities performed" is a definition of
 1. Record
 2. Document
 3. Evidence
 4. None of the above
38. Quality Management System consists of
 1. Processes
 2. Documents
 3. Records
 4. All of the above
39. Which one of the following is not a type of testing
 1. Alfa Testing
 2. Beta Testing
 3. Gamma Testing
 4. White Box Testing
40. "Testing of an application when development is nearing completion; minor design changes may still be made as a result of such testing. Typically done by end-users or others, not by programmers or testers" is a definition of
 1. Alfa Testing
 2. Beta Testing
 3. Gamma Testing
 4. White Box Testing
41. "Testing when development and testing are essentially completed and final bugs and problems need to be found before final release. Typically done by end-users or others, not by programmers or testers." is a definition of
 1. Alfa Testing
 2. Beta Testing
 3. Gamma Testing
 4. Black Box Testing
42. "Not based on any knowledge of internal design or code. Tests are based on requirements and functionality." Is a definition of
 1. Unit Testing
 2. Beta Testing
 3. White Box Testing
 4. Black Box Testing
43. "Based on knowledge of the internal logic of an application's code. Tests are based on coverage of code statements, branches, paths, conditions"
 1. Unit Testing
 2. Beta Testing
 3. White Box Testing
 4. Black Box Testing
44. Which of the following is True for "Software Testing"
 1. Can compensate for poor Software project
 2. An indicator of Absence of Defects
 3. Give "Defect Free Software"
 4. Adds confidence to the development team
45. Which of the following are SDLC process models
 1. Waterfall
 2. Evolutionary
 3. Spiral
 4. All of the above
46. PDCA cycle is
 1. Plan- Do – Check - Act
 2. Plan – Develop – Check - Act
 3. Plan – Develop - Configure - Act
 4. None of the above
47. Verification means
 1. Are you meeting the specified requirements?
 2. Are you building the product right?
 3. None of the above
 4. Both 1 and 2
48. Validation means
 1. Are you meeting the operational need?
 2. Does this product meet its intended use in the intended environment?
 3. Are you building the right product?
 4. All of the above
49. Debugging is a consequence of
 1. analysis
 2. design
 3. testing
 4. maintenance
50. Which is the non-technical factor of maintenance cost?
 1. program age
 2. programming style
 3. program validation
 4. None of the above